

Internship Project Report

By Parnika Gaura
(Intern)

Introduction (自己紹介)

Konnichiwa!

Hajimemashite.

Parnika desu.

Watashi wa Nihon de intānshippu o shimashita.

Kyou wa watashi no intānshippu to purojekuto ni tsuite happyou shimasu.

Yoroshiku onegaishimasu.

Contents

1. What I did in my Internship?
2. What I did in Japan?

Internship Overview

- ❑ Internship in Japan
- ❑ Worked on Zebra Crossing Detection
- ❑ Used RGB camera and Machine Learning



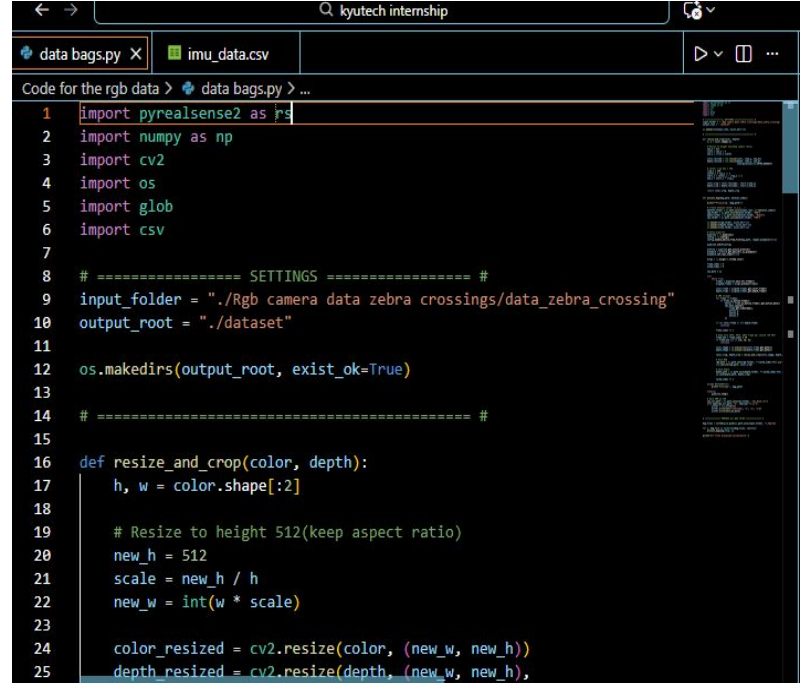
Recorded Video using RGB Camera

- Used RGB camera to record zebra crossings
- Recorded videos in real-world road conditions
- Ensured stable camera positioning
- Captured clear visibility of road markings



Converted Video into Images

- Extracted frames from recorded videos
- Converted video into multiple image frames
- Selected specific frames per second
- Saved images in structured folders

A screenshot of a code editor window with a dark theme. The title bar shows 'kyutech internship'. There are two tabs: 'data bags.py' and 'imu_data.csv'. The main editor area shows Python code for processing video frames. The code includes imports for pyrealsense2, numpy, cv2, os, glob, and csv. It defines settings for input and output folders, creates the output directory, and defines a function to resize and crop video frames to a height of 512 pixels.

```
1 import pyrealsense2 as rs
2 import numpy as np
3 import cv2
4 import os
5 import glob
6 import csv
7
8 # ===== SETTINGS ===== #
9 input_folder = "../Rgb camera data zebra crossings/data_zebra_crossing"
10 output_root = "./dataset"
11
12 os.makedirs(output_root, exist_ok=True)
13
14 # ===== #
15
16 def resize_and_crop(color, depth):
17     h, w = color.shape[:2]
18
19     # Resize to height 512(keep aspect ratio)
20     new_h = 512
21     scale = new_h / h
22     new_w = int(w * scale)
23
24     color_resized = cv2.resize(color, (new_w, new_h))
25     depth_resized = cv2.resize(depth, (new_w, new_h),
```

(code for converting videos to image)

Image Preprocessing

- Resized images from raw images
- Cropped to required dimensions (height=512 & width=216)
- Organized dataset
- Prepared images for annotation



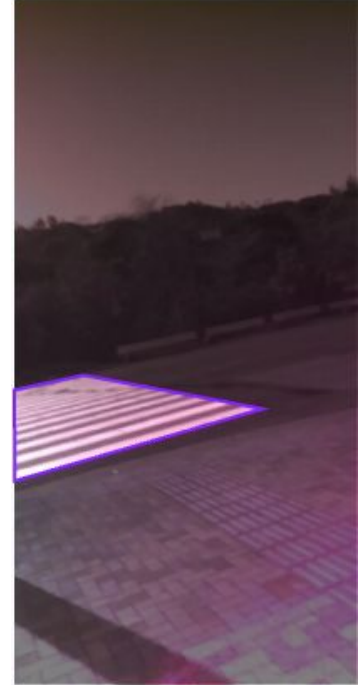
Raw image



Resized image

Image Annotation

1. Annotated 2611 images with the tool(Roboflow)
2. Drew bounding boxes around zebra crossings
3. Labeled each object
 - a. No-Light or Red or Green
 - b. No-Zebra or Zebra's Bounding Box
 - c. Depth maps(from Realsense)
4. Created training and testing datasets



(Zebra crossing with
bounding box)

Machine Learning Model used

- Multiple deep learning models were used to evaluate zebra crossing detection performance.
- The models tested include **MobileNetV3**, **LYTNetV2**, and **EfficientNet-B0**.
- Each model was evaluated in **First Generation and Second Generation versions**.
- RGB images and depth information were used to improve detection accuracy

Models used & Output

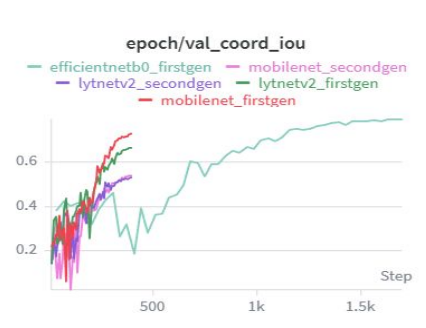
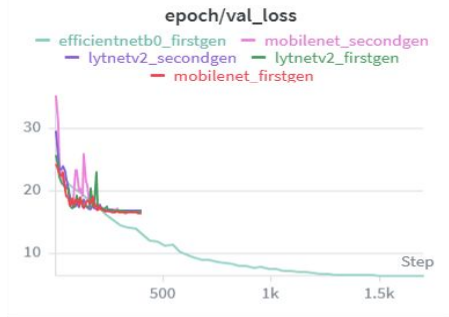
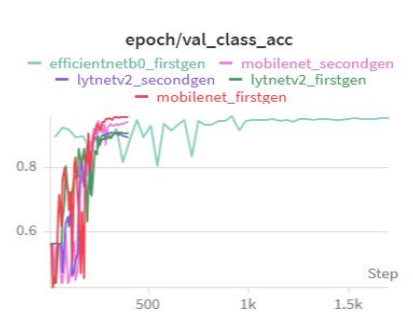
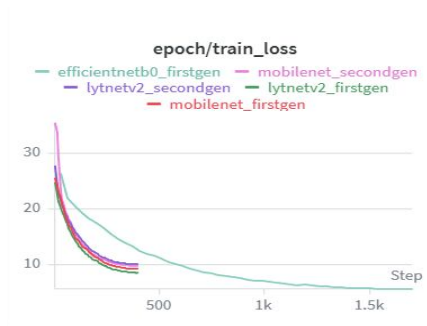
Models used:

1. MobileNetV3 – FirstGen
2. MobileNetV3 – SecondGen
3. LYTNetV2 – FirstGen
4. LYTNetV2 – SecondGen
5. EfficientNet-B0 – FirstGen

Model results:

- ❑ **Blue box:** Ground Truth (Actual Zebra Crossing)
- ❑ **Green box:** Model Prediction
- ❑ **Red box:** Depth Region of Interest

Training and Validation Performance



- Training loss decreases steadily, indicating stable learning.
- Validation accuracy improves across epochs.
- IoU score shows improvement in bounding box localization.

Final Results

Model name :

MobileNetV3-S-SecondGen[MobileNetV3-S-FirstGen] => Val Loss: 9.9870 | Coord IoU: 0.7262 | Class Acc: 0.9565

[MobileNetV3-S-SecondGen] => Val Loss: 9.9914 | Coord IoU: 0.5349 | Class Acc: 0.9420

[LYTNet-V2-FirstGen] => Val Loss: 9.4137 | Coord IoU: 0.6589 | Class Acc: 0.9040

[LYTNet-V2-SecondGen] => Val Loss: 10.4528 | Coord IoU: 0.5223 | Class Acc: 0.8949

[EfficientNet-Bo-FirstGen] => Val Loss: 6.0721 | Coord IoU: 0.7889 | Class Acc: 0.9565

Model name : MobileNetV3-S-FirstGen

Total Params : 5,218,840

Total FLOPs : 0.25 GFLOPs

Model name : LYTNet-V2-FirstGen

Total Params : 7,376,040

Total FLOPs : 0.16 GFLOPs

Model name : EfficientNet-B0-FirstGen

Total Params : 25,180,676

Total FLOPs : 1.44 GFLOPs

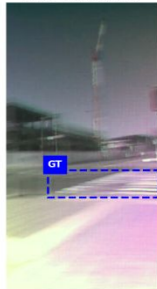
Example

Sample 1 | Blue = Ground Truth | Green = Predictions | Red = Depth ROI

Ground Truth
GT Class: Zebra crossing



MobileNetV3-S-FirstGen
Pred : No Object



MobileNetV3-S-SecondGen
Pred : Zebra crossing



LYTNet-V2-FirstGen
Pred : No Object



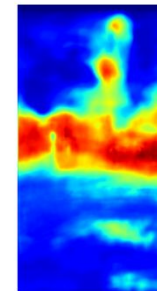
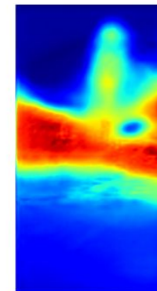
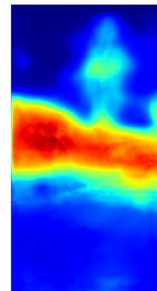
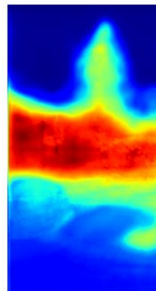
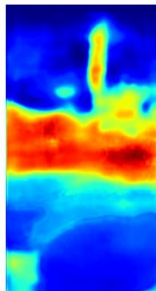
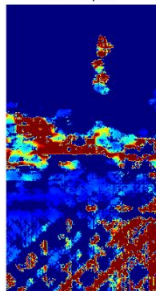
LYTNet-V2-SecondGen
Pred : Zebra crossing



EfficientNet-B0-FirstGen
Pred : Zebra crossing



GT Depth



Final Results & Conclusion

- Successfully collected **RGB, depth, and IMU data** for zebra crossing detection.
- Converted videos into images and **annotated bounding boxes** for training.
- Trained and evaluated **five deep learning models**.
- Models using **depth information showed improved detection performance**.
- The system demonstrates potential for **real-world pedestrian safety applications**.

Experience in Japan

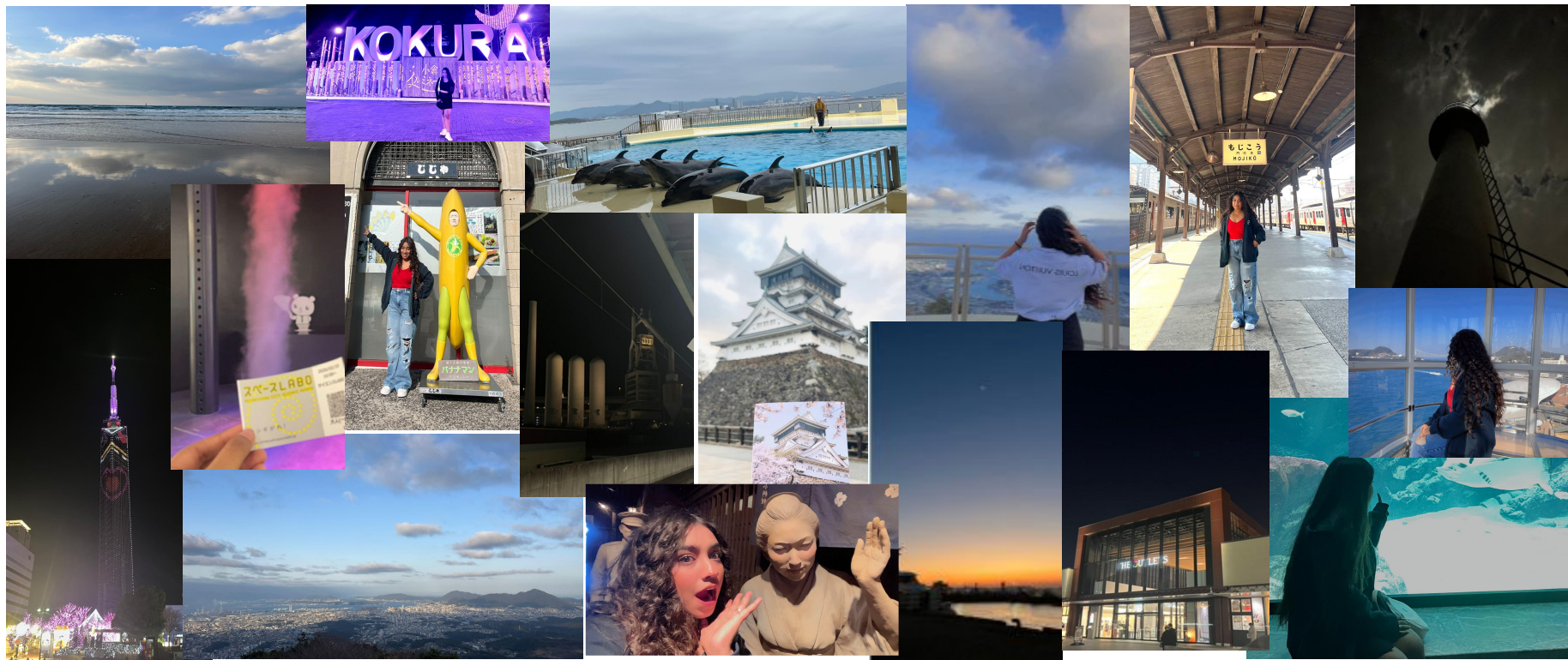
Internship Experience in Japan

- Worked in professional research environment
- Learned Japanese work culture
- Improved teamwork and communication skills
- Gained international exposure

Places I Explored in Japan

- **Kokura Castle** – Historic Japanese castle and cultural landmark in Kitakyushu
- **Marine World** – Aquarium showcasing marine life and ocean ecosystems
- **Mount Sarakura** – Famous viewpoint with panoramic night view of the city
- **Iwaya Beach** – Relaxing coastal area with beautiful sea scenery
- **Cape Tomigahana** – Scenic coastal spot known for peaceful ocean views
- **Space World Area & Space LABO Museum** – Science and space-themed learning experience
- **The OUTLETS Kitakyushu** – Large shopping complex with international brands
- **Fukuoka Tower** – Tallest seaside tower in Japan with city views
- **Tenjin & Hakata** – Popular districts for shopping, food, and urban culture
- **Mojiko & Karato Market** – Historic port area and famous seafood market
- **Amusement park-** Had the ferry wheel ride

Places I visited in Japan



Japanese Culture and Lifestyle

- Experienced Japanese **food culture** (sushi, matcha, desserts)
- Observed **cleanliness and discipline** in public places
- Efficient **public transportation system** (trains,buses)
- Visited temples
- Learned basic **Japanese communication**
- Experienced **Cherry Blossom**





THANK YOU

Arigatou Gozaimasu (ありがとうございます)

Thank you for your time and attention.

I truly appreciate the opportunity to present my internship experience
in Japan.

Presented by -Parnika Gaura
Intern at Kyutech University
Japan
